

RidePulse Analytics — Complete Analytical Report

SQL-Based Ride Booking Intelligence System

This project is a production-style mobility analytics solution built using PostgreSQL with advanced SQL concepts, business KPIs, forecasting logic, and operational intelligence queries. The project analyzes ride-booking operations using transactional, behavioral, payment, weather, and driver-performance datasets.

Data Architecture

The project uses six integrated datasets:

Table	Purpose
trips	Core transactional ride data
customers	Customer demographics & retention
drivers	Driver performance analytics
payments	Gateway & transaction monitoring
complaints	Service quality analysis
weather_traffic	External operational impact analysis

The schema supports:

- relational joins
 - operational KPIs
 - forecasting
 - behavioral analysis
 - dynamic pricing intelligence
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SQL Concepts Used

This project demonstrates strong command over advanced PostgreSQL analytics:

Advanced SQL Features

- CTEs (`WITH`)
- Window Functions
- `ROW_NUMBER()`

- `DENSE_RANK()`
 - `LAG()`
 - Running totals
 - Moving averages
 - Conditional aggregation
 - `FILTER()`
 - Date extraction
 - Forecasting logic
 - Churn analysis
 - Cohort-style trend tracking
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Business Analytics Modules

1. Revenue Analytics

Queries analyze:

- city-wise revenue
- monthly growth
- cancellation loss
- weather-based profitability
- route profitability

Key Insights

- Revenue uses:

```
fare_amount * surge_multiplier
```

which simulates realistic ride-booking pricing behavior.

- Running revenue totals help monitor cumulative business growth.
- Revenue trend analysis supports executive dashboarding.

Relevant Queries:

- Total Revenue Per City
 - Monthly Revenue Trend
 - Running Revenue Total
 - Revenue by Weather
 - MTD Growth
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Driver Performance Intelligence

The project contains sophisticated driver-level analysis.

Metrics Measured

- total trips
- total earnings
- driver efficiency
- cancellations
- complaint counts
- top-performing drivers

Advanced Logic

A weighted “Driver Efficiency Score” combines:

- ratings
- completed rides
- earnings

This resembles real-world mobility platform scoring systems.

Fraud Detection

The project identifies:

- abnormal cancellation rates
- suspicious drivers
- operational inefficiencies

Relevant Queries:

- Top Earning Drivers
- Driver Efficiency Score
- Fraudulent Cancellation Patterns
- Top 5 Drivers by City



Customer Intelligence & Retention

The project performs advanced customer analytics.

Key Features

- repeat customer analysis
- customer lifetime value
- churn prediction

- cancellation behavior

Churn Prediction

Inactive customers (>60 days) are identified using:

```
CURRENT_DATE - last_booking
```

This mirrors real SaaS/ride-booking retention analytics.

Relevant Queries:

- Repeat Customers
- Customer Lifetime Value
- Churn Prediction



Payment & Financial Risk Analytics

The project evaluates:

- payment gateway reliability
- payment failure rate
- transaction behavior

Key Strength

Uses PostgreSQL `FILTER()` aggregation elegantly for failed-payment calculations.

Example business use:

- gateway monitoring
- fintech stability analysis
- transaction reliability KPIs

Relevant Queries:

- Most Used Payment Gateway
- Payment Failure Rate



Weather & Dynamic Pricing Analytics

A standout feature of the project is operational intelligence using external conditions.

Analyzed Factors

- rain impact
- surge pricing
- traffic conditions
- demand fluctuation

Dynamic Surge Logic

The project categorizes:

- low surge
- medium surge
- high surge

This closely resembles Uber/Rapido/Ola dynamic pricing systems.

Relevant Queries:

- Weather Impact on Demand
- Rain vs Non-Rain Profitability
- Highest Surge Days
- Dynamic Surge Revenue Impact



Forecasting & Predictive Analytics

This project goes beyond basic SQL reporting.

Forecasting Implemented

- moving average demand estimation
- monthly growth analysis
- next-month ride prediction

Window Function Usage

Uses:

- LAG ()
- rolling comparisons
- month-over-month growth

This demonstrates intermediate-to-advanced analytical engineering capability.

Relevant Queries:

- City Demand Forecasting
 - MTD Growth
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Operational Intelligence

The project identifies:

- peak booking hours
- profitable routes
- ride duration trends
- booking density

Most Profitable Route Pair

One of the strongest business queries:

- computes fare/km efficiency
- ranks top routes city-wise

This is highly relevant for:

- mobility optimization
 - pricing strategy
 - driver allocation
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Strongest Technical Highlights

Excellent Use of:

- analytical SQL
- business KPIs
- operational intelligence
- forecasting logic
- ranking functions
- real-world ride-booking metrics

Particularly Impressive Sections

- Driver Efficiency Score
- Fraud Detection Logic
- Dynamic Surge Analytics
- Route Profitability

- Churn Prediction
- Forecasting Query

Areas for Improvement

1. Some GROUP BY clauses unnecessary

A few window-function queries contain redundant `GROUP BY`.

Example:

- forecasting query
- growth-rate query

2. Weekend Logic Bug

This section:

```
WHEN EXTRACT(DOW FROM booking_date) IN (0,5)
```

incorrectly marks Friday as weekend.

Correct:

```
IN (0,6)
```

for Sunday + Saturday.

3. Revenue Trend Sorting

Monthly revenue query sorts by revenue instead of chronological month order.

Could use:

```
ORDER BY DATE_TRUNC('month', booking_time)
```

4. Forecasting Logic Can Improve

Current prediction uses average growth rate.

Advanced future improvement:

- exponential smoothing
- ARIMA
- Prophet
- regression forecasting

Overall Project Evaluation

Category	Rating
SQL Complexity	★ ★ ★ ★ ★
Business Understanding	★ ★ ★ ★ ★
Real-World Relevance	★ ★ ★ ★ ★
Analytical Thinking	★ ★ ★ ★ ★
Dashboard Readiness	★ ★ ★ ★ ★
Data Modeling	★ ★ ★ ★
Predictive Analytics	★ ★ ★ ★